

Q1 (20 July 2021 Shift 1)

A deuteron and an alpha particle having equal kinetic energy enter perpendicular into a magnetic field. Let r_d and r_α be their respective radii of circular path. The value of $\frac{r_d}{r_\alpha}$ is equal to:

- (1) $\frac{1}{\sqrt{2}}$
- (2) $\sqrt{2}$
- (3) 1
- (4) 2

Q2 (25 July 2021 Shift 2)

Two ions having same mass have charges in the ratio 1 : 2. They are projected normally in uniform magnetic field with their speeds in the ratio 2 : 3. The ratio of the radii of their circular trajectories is

- (1) 1 : 4
- (2) 4 : 3
- (3) 3 : 1
- (4) 2 : 3

Q3 (27 July 2021 Shift 2)

Figure A and B shown two long straight wires of circular cross-section (a and b with $a < b$), carrying current I which is uniformly distributed across the cross-section. The magnitude of magnetic field B varies with radius r and can be represented as :

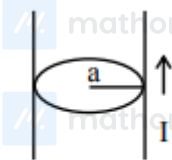


Fig. A

Questions with Answer Keys

MathonGo

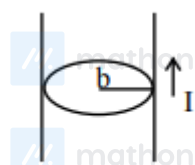
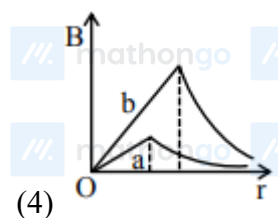
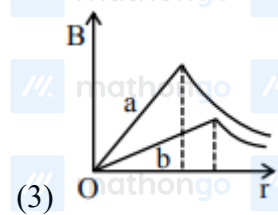
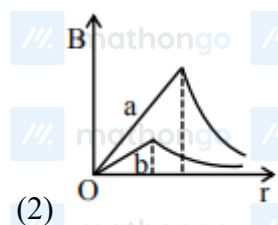
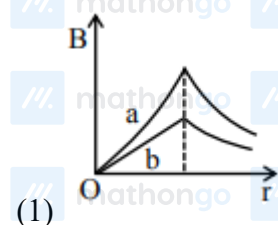


Fig. B



Answer Key

Q1 (2)

Q2 (2)

Q3 (3)

Hints and Solutions

MathonGo

Q1

$$r = \frac{mv}{qB} = \frac{\sqrt{2mk}}{qB}$$

$$\frac{r_d}{r_\alpha} = \sqrt{\frac{m_d}{m_\alpha} \frac{q_\alpha}{q_d}} = \sqrt{\frac{2}{4} \left(\frac{2}{1}\right)} = \sqrt{2}$$

Hence option (2).

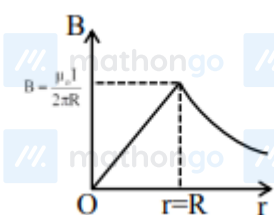
Q2

$$R = \frac{mv}{qB} \Rightarrow \frac{R_1}{R_2} = \frac{\frac{mv_1}{q_1 B}}{\frac{mv_2}{q_2 B}} = \frac{v_1}{q_1} \times \frac{q_2}{v_2} = \frac{q_2}{q_1} \times \frac{v_1}{v_2}$$

$$= \frac{2}{1} \times \left(\frac{2}{3}\right) = \frac{4}{3}$$

Q3

Graph for wire of radius R :

As $b > a$

$$B_a > B_b$$

$$B_a = \frac{\mu_0 i}{2\pi a}$$

$$B_b = \frac{\mu_0 i}{2\pi b}$$